

Machine Learning-Based Monitoring and Detection of Persistent Changes in Wetlands: A Case Study of Kigali, Rwanda

BSc. in Software Engineering

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CHAPTER ONE: INTRODUCTION

1.1 Background to the Study

Wetlands are important ecosystems that provide environmental and socio-economic benefits. They help regulate water, reduce flood risks, protect water quality, support biodiversity, store carbon, and support surrounding communities (REMA, 2021). In Rwanda, these functions are increasingly important as wetlands face pressure from human activities and land-use change.

Rwanda's wetland coverage declined from 276,488 hectares in 1988 to 176,337 hectares in 2016, representing a 36% reduction over 28 years (REMA, 2021). In Kigali, the wetland network declined from approximately 100 km² in 2013 to 72 km², with industrial development and urban agriculture contributing to the loss (REMA, 2020). Major pressures on wetlands include agriculture, urbanization, infrastructure development, erosion, and other human activities (REMA, 2021).

Rwanda has responded through field assessments, GIS, satellite-based monitoring, and automated remote-sensing approaches. For example, the Kigali Urban Wetland and Restoration Monitoring System uses satellite imagery and deep-learning techniques to assess wetland habitat extent and human impacts (ESA, 2023). However, national wetland monitoring has generally been structured around periodic assessments, with monitoring frequency influenced by available resources and management objectives (REMA, 2021). This highlights the need for further research into how changes in wetland conditions can be identified as they develop over time.

1.2 Statement of the Problem

Rwanda's wetlands are under increasing pressure from agriculture, urbanization, infrastructure development, and other human activities. These pressures have contributed to wetland degradation and reduced their ability to provide important ecosystem services (REMA, 2021).

Although Rwanda uses satellite imagery, GIS, field assessments, and automated remote-sensing approaches for wetland monitoring, existing systems mainly focus on mapping wetland extent and assessing changes over defined periods. For example, the Kigali Urban Wetland and Restoration Monitoring System provides multi-year habitat-change products, including a 2017–2021 assessment (ESA, 2023).

Another challenge is identifying whether changes observed in wetland conditions are temporary or persistent. Existing research has demonstrated the use of multi-temporal satellite data for monitoring wetland conditions, but there is still a need for approaches that

can identify persistent deviations from normal wetland patterns across successive observations (Dabrowska-Zielinska et al., 2023).

To address these problems, this study introduces an ML-based monitoring system that will analyse multi-temporal Sentinel-2 imagery, learn normal wetland patterns, detect persistent deviations, and quantify and map affected areas. The system will be evaluated using measurable metrics to determine its effectiveness in identifying persistent changes in wetland conditions.

1.3 Aim and Objectives of the Study

Aim

To develop an ML-based wetland monitoring system that detects persistent changes in wetland conditions using multi-temporal Sentinel-2 satellite imagery to support timely identification of emerging wetland degradation.

1.3.1 List of the specific objectives

Break down the overall aim into 3 specific objectives:

1. To review existing wetland monitoring approaches and analyse multi-temporal Sentinel-2 data to identify patterns and indicators of persistent wetland changes.
2. To develop an ML-based monitoring system that learns normal wetland patterns and detects, quantifies, and maps persistent deviations from those patterns.
3. To evaluate the developed system using technical and problem-specific metrics to determine its effectiveness in identifying persistent changes in wetland conditions.

1.4 Research questions

1. What patterns and indicators in multi-temporal Sentinel-2 data can be used to identify persistent changes in wetland conditions?
2. How can an ML-based system be developed to detect, quantify, and map persistent deviations from normal wetland patterns?
3. How effective is the developed system in identifying persistent changes in wetland conditions based on defined technical and problem-specific metrics?

1.5 Project scope

Geographically, the study will focus on one selected wetland within Kigali City rather than the entire country. The study will use multi-temporal Sentinel-2 satellite imagery of the selected wetland to develop and evaluate the proposed monitoring system.

Technically, the project will focus on detecting persistent changes in wetland conditions, quantifying affected areas, and producing maps of detected changes. The system will be developed as a prototype and evaluated using historical satellite observations and available reference data.

The study will not cover all wetlands in Rwanda, all forms of wetland degradation, or nationwide deployment. Its findings will therefore demonstrate the feasibility of the proposed approach within the selected study area and provide a basis for future expansion to other wetlands.

1.6 Significance and Justification

The successful implementation of the system will provide a more consistent way of identifying persistent changes in wetland conditions using regularly available satellite imagery. This can support earlier identification of areas that may require further environmental assessment.

The system will also provide quantified and mapped information on affected wetland areas, which can support environmental monitoring and help authorities prioritize areas for further investigation and intervention.

For research and future development, the project will demonstrate the application of machine learning and multi-temporal satellite data to wetland monitoring in Rwanda. The developed approach can also provide a foundation for future expansion to other wetlands.

1.7 Research Budget

The project will primarily use freely available satellite imagery and open-source software tools. Therefore, the research budget is limited to essential expenses that may be required for data collection, field verification, and internet access.

Item	**Description**	**Cost Estimated (RWF)**
Internet/Data	Downloading satellite imagery, accessing research literature, and development	40,000
Field visits	Transport for site observation and verification of selected wetland areas	50,000
Printing and Documentation	Printing research and field materials	20,000
Total		**110,000**

1.8 Research Timeline

Activity	**Month 1**	**Month 2**	**Month 3**
Literature review and problem analysis	✓		

Dataset collection and preparation	✓	✓	
System and model design	✓		
Model development and training		✓	
System implementation		✓	✓
Model and system testing			✓
Results collection and evaluation			✓
Documentation and final report		✓	✓
Final presentation and submission			✓

1.9 Definition of Terms

Wetland: An area of land that is permanently or seasonally saturated with water and supports vegetation and ecological processes adapted to wet conditions.

Multi-temporal Satellite Imagery: Satellite images of the same geographical area collected at different points in time, allowing changes in environmental conditions to be analysed.

Sentinel-2: A European Space Agency satellite mission that provides multispectral Earth observation imagery suitable for monitoring land and environmental conditions.

Machine Learning (ML): A branch of artificial intelligence in which computer systems learn patterns from data and use those patterns to make predictions or identify changes.

Persistent Change: A change in wetland conditions that continues across successive observations rather than appearing as a temporary variation.

Anomaly: A significant deviation from the normal or expected pattern of wetland conditions identified by the monitoring system.

Wetland Degradation: A decline in the ecological condition or functionality of a wetland caused by factors such as human activities or environmental changes.

Change Detection: The process of identifying and measuring differences in the condition of a geographical area across different time periods using satellite or other spatial data.

CHAPTER TWO: LITERATURE REVIEW

2.1 Introduction

The literature review focused on software-related research in **machine learning, remote sensing, satellite image analysis, multi-temporal change detection, and automated wetland monitoring**, particularly studies using Sentinel-2 imagery. A structured literature search was conducted using indexed platforms including **Scopus, Web of Science, ScienceDirect, IEEE Xplore, SpringerLink, and MDPI**, using keywords such as *wetland monitoring, machine learning, Sentinel-2, remote sensing,* and *change detection*. The identified papers were screened based on their relevance, publication date, use of computational methods, and connection to wetland monitoring. A sample of the most relevant papers was selected for detailed analysis of their methods, datasets, and limitations.

2.2 **Conceptual Review**(including the key concepts and themes in both dependent and independent variables)

This section reviews the key concepts related to the proposed ML-based wetland monitoring system, focusing on the independent variables and their relationship with the dependent variable.

2.2.1 Machine Learning

Machine learning (ML) enables computer systems to learn patterns from data and use them to identify or predict changes. In this study, ML will be used to analyse satellite data and identify patterns in wetland conditions.

2.2.2 Multi-temporal Sentinel-2 Imagery

Multi-temporal imagery consists of satellite images of the same area collected at different times. Sentinel-2 provides multispectral data that can be used to monitor changes in vegetation, water, and land characteristics. In this study, it will provide the main input data for the monitoring system.

2.2.3 Change Detection

Change detection is the process of identifying differences between observations collected at different times. In wetland monitoring, it can help identify changes in vegetation, water extent, and other environmental characteristics.

2.2.4 Persistent Change

Persistent change refers to a change that continues across successive observations rather than occurring temporarily. This study focuses on persistent changes to distinguish potential degradation from short-term variations.

2.2.5 Wetland Condition

Wetland condition describes the state of a wetland based on measurable characteristics such as vegetation and water-related indicators. In this study, changes in wetland condition represent the main outcome to be detected, quantified, and mapped.

Overall, the concepts are connected as follows: **multi-temporal Sentinel-2 imagery provides the data, machine learning analyses the data, and change-detection methods identify persistent changes in wetland conditions.**

2.3 Historical Background of the Research Topic

Wetland monitoring has gradually developed from field-based assessment to satellite remote sensing and, more recently, machine learning-based analysis.

2.3.1 Origins and Early Context of the Research Problem

Early wetland monitoring mainly relied on field surveys and direct observation, which were limited in coverage and frequency. The introduction of remote sensing enabled wetlands to be monitored over larger areas and across different periods.

2.3.2 Key Milestones

The development of multispectral satellite missions, including Sentinel-2, improved the availability and quality of data for environmental monitoring. Machine learning has further enabled automated classification, mapping, and change detection from satellite imagery.

2.3.3 Historical Gaps and Transition to the Current Study

Existing approaches have largely focused on mapping wetland extent and detecting changes between time periods. This study builds on these developments by focusing on the detection of **persistent changes** in wetland conditions using multi-temporal Sentinel-2 imagery and machine learning.

2.4 Theoretical Review (Concluding with a theoretical framework - Where necessary)

2.4.1 Foundational Theories

Remote Sensing Theory explains how information about the Earth's surface can be obtained from satellite sensors without direct physical contact. It provides the basis for using Sentinel-2 imagery to observe and monitor changes in wetland conditions.

Change Detection Theory provides the basis for identifying differences in the same geographical area across different time periods. In this study, it supports the identification of persistent changes in wetland conditions from multi-temporal satellite imagery.

Machine Learning Theory explains how computational models can learn patterns from data and use them to identify new or abnormal patterns. It supports the development of the proposed system for learning normal wetland conditions and detecting persistent deviations.

2.5 Overview of Existing System (Review of related technical solutions)

Existing software systems such as GlobWetland II, GreenEye, and WetSAT-ML use satellite imagery, GIS, and machine learning to support wetland monitoring. However, they mainly focus on mapping, water monitoring, or general change detection rather than identifying persistent changes in wetland conditions.

2.5.1 Strengths and Weaknesses of Existing Systems

These systems support automated monitoring and analysis of satellite data, but they have limited focus on distinguishing persistent changes from temporary variations. This creates an opportunity for the proposed system to use multi-temporal Sentinel-2 imagery and machine learning to detect, quantify, and map persistent wetland changes.

2.6 Review of Related Works (Empirical Review)

Previous studies have applied machine learning and satellite imagery to wetland mapping and change detection. Studies using Sentinel-2 have demonstrated its usefulness for monitoring wetland vegetation, water extent, and land-cover changes. Recent research has also applied Random Forest and deep learning models to improve automated wetland classification and change detection.

2.6.1 Summary of Reviewed Literature

Previous studies have demonstrated the use of machine learning and satellite imagery for automated wetland monitoring and change analysis. Bhatnagar et al. (2021), for example, combined Sentinel-2 satellite imagery with drone imagery and Random Forest classification to produce seasonal maps of wetland vegetation communities in Ireland. Similarly, Mugiraneza et al. (2022) used Sentinel-2 MSI data to monitor urbanization and environmental impacts in Kigali, Rwanda. At a broader level, Jafarzadeh et al. (2022) reviewed 344 studies on remote sensing and machine learning for wetland monitoring, showing that satellite imagery and machine learning have been widely applied to wetland classification and monitoring. More recent work has extended these approaches to temporal change analysis. Dodangeh and Shah-Hosseini (2026) used Sentinel-2 time-series data and deep learning to quantify water-area changes in the Miankaleh Wetland, while Dzomeku et al. (2026) used multi-temporal Sentinel-2 imagery and Random Forest classification to analyse land-cover changes in Ghana's Keta Lagoon Complex. These studies demonstrate the usefulness of satellite imagery and machine learning for wetland mapping and change detection; however, their main focus is on classification, extent mapping, water-area changes, or land-cover transitions rather than identifying whether observed changes persist across successive observations.

2.7 General Comments

The reviewed literature shows that remote sensing, multi-temporal satellite imagery, and machine learning provide effective tools for automated wetland monitoring. However, limited attention has been given to identifying persistent changes rather than temporary variations. The proposed study addresses this gap by developing an ML-based system that uses multi-temporal Sentinel-2 imagery to detect, quantify, and map persistent changes in wetland conditions.

CHAPTER THREE: SYSTEM ANALYSIS AND DESIGN

3.1 Introduction

This chapter presents the methodology and system design approaches used to develop the proposed ML-based wetland monitoring system. The study will use a **design science research approach** to guide the development and evaluation of the system. The system will be designed using **machine learning and multi-temporal Sentinel-2 satellite data** to detect, quantify, and map persistent changes in wetland conditions. The development process will include data collection and preparation, model development, system implementation, testing, and evaluation.

3.2 Research Design

The study will use a **Design Science Research (DSR)** approach combined with an iterative Software Development Life Cycle (SDLC) to develop and evaluate an ML-based wetland monitoring system. The research will follow five main stages: data selection and preparation, baseline development, machine learning model development, persistent-change detection, and system evaluation. The approach is designed to determine whether multi-temporal Sentinel-2 imagery and machine learning can be used to identify persistent changes in the condition of a selected wetland in Kigali.

3.2.1 Dataset and Dataset Selection

The study will use multi-temporal **Sentinel-2 satellite imagery** covering one selected wetland in Kigali. Historical observations will be collected over multiple years to provide sufficient temporal information for identifying normal seasonal patterns and longer-term changes.

The selected imagery will be filtered based on spatial coverage, acquisition date, image quality, and cloud conditions. Relevant Sentinel-2 spectral bands and derived indices, particularly vegetation- and water-related indicators, will be used as input features. The data will be processed into consistent observations for the selected study area so that conditions can be compared across time.

The selected wetland will be chosen based on its relevance to the research problem, availability of suitable Sentinel-2 observations, and availability of reference information that can support evaluation of detected changes.

3.2.2 Data Preprocessing and Feature Extraction

The satellite imagery will undergo preprocessing before being used by the machine learning model. This will include cloud and unsuitable-pixel filtering, spatial clipping to the selected wetland, and preparation of comparable observations across different dates.

Relevant spectral bands and indices will then be extracted to represent changes in vegetation, water, and other observable wetland characteristics. The resulting time-series dataset will contain observations of the wetland condition at different points in time and will be used to establish normal temporal patterns.

3.2.3 Baseline Model

A conventional change-detection method will be developed as a baseline for comparison with the proposed machine learning approach. The baseline will compare wetland indicators between observations and identify areas where measurable changes occur.

The purpose of the baseline is to establish how a simpler conventional approach performs before introducing the proposed machine learning method. This comparison will help determine whether the proposed approach provides improved identification of persistent changes rather than only detecting differences between individual observations.

3.2.4 Proposed Machine Learning Model

The proposed model will learn the normal temporal patterns of the selected wetland from historical Sentinel-2 observations. The model will analyse changes in the extracted features and identify observations that significantly deviate from the established normal patterns.

A change will be considered **persistent** when a detected deviation continues across successive observations rather than occurring in only one observation. The persistence criterion will therefore be applied to the model's detected deviations to distinguish sustained changes from short-term or temporary variations.

The system will then quantify the spatial extent of areas identified as having persistent changes and generate maps showing their geographic distribution. The detected changes will be compared with available reference information to assess whether the system is correctly identifying areas of change.

3.2.5 Training, Validation, and Testing

The available historical observations and reference data will be divided into training, validation, and testing components where sufficient data are available. Training data will be used to develop the model's understanding of normal wetland patterns, validation data will support model and threshold selection, and testing data will be reserved for final performance evaluation.

The testing stage will use observations that were not used during model development to provide an independent assessment of the system's ability to identify persistent changes.

3.2.6 Evaluation Metrics

The proposed system will be evaluated using both classification and spatial accuracy measures. **Precision, recall, and F1-score** will be used to assess the ability of the system to correctly identify persistent changes. **Accuracy** will be used where the evaluation task is formulated as a classification problem, while **Intersection over Union (IoU)** will be used to measure the spatial overlap between detected change areas and available reference areas.

The proposed machine learning approach will also be compared with the baseline method. The comparison will determine whether the proposed approach provides more effective identification and spatial mapping of persistent wetland changes.

3.3 System Analysis

3.3.1 Functional Requirements of the System

The system shall:

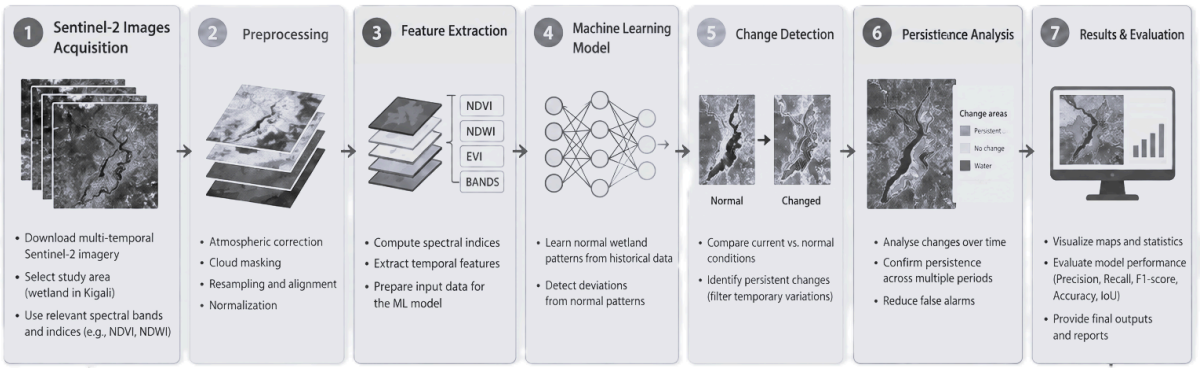
1. Acquire and process multi-temporal Sentinel-2 satellite imagery.
2. Preprocess and prepare the imagery for analysis.
3. Analyse wetland patterns using the machine learning model.
4. Detect persistent changes in wetland conditions.
5. Quantify the affected areas.
6. Generate maps and visual outputs showing detected changes.
7. Provide evaluation results for the developed model.

3.3.2 Non-functional Requirements of the System

The system shall be:

- **Accurate:** Produce reliable change-detection results.
- **Efficient:** Process satellite imagery within reasonable time.
- **Usable:** Present results through a clear and understandable interface.
- **Scalable:** Allow the approach to be extended to other wetland areas.
- **Maintainable:** Use a modular structure that allows future improvements to the model and system.

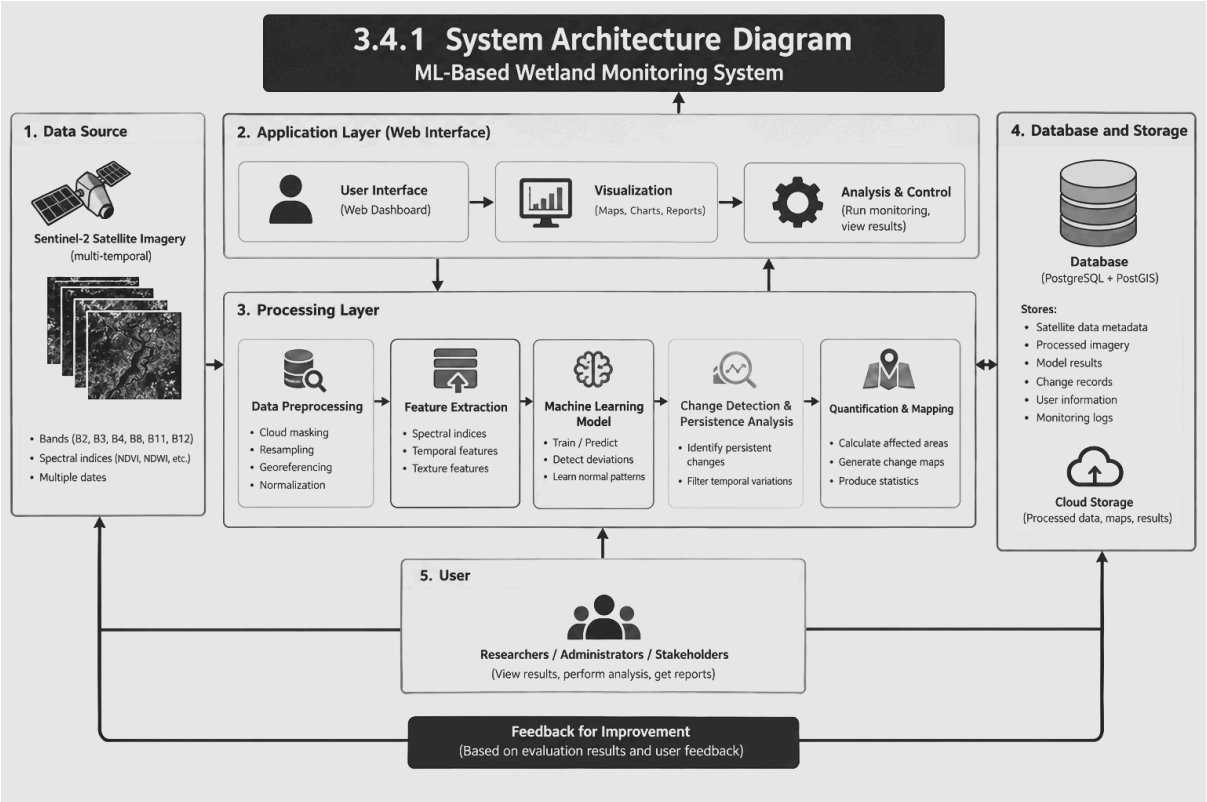
3.3.3 Proposed Model Diagram customized to your mission



3.4 System Design

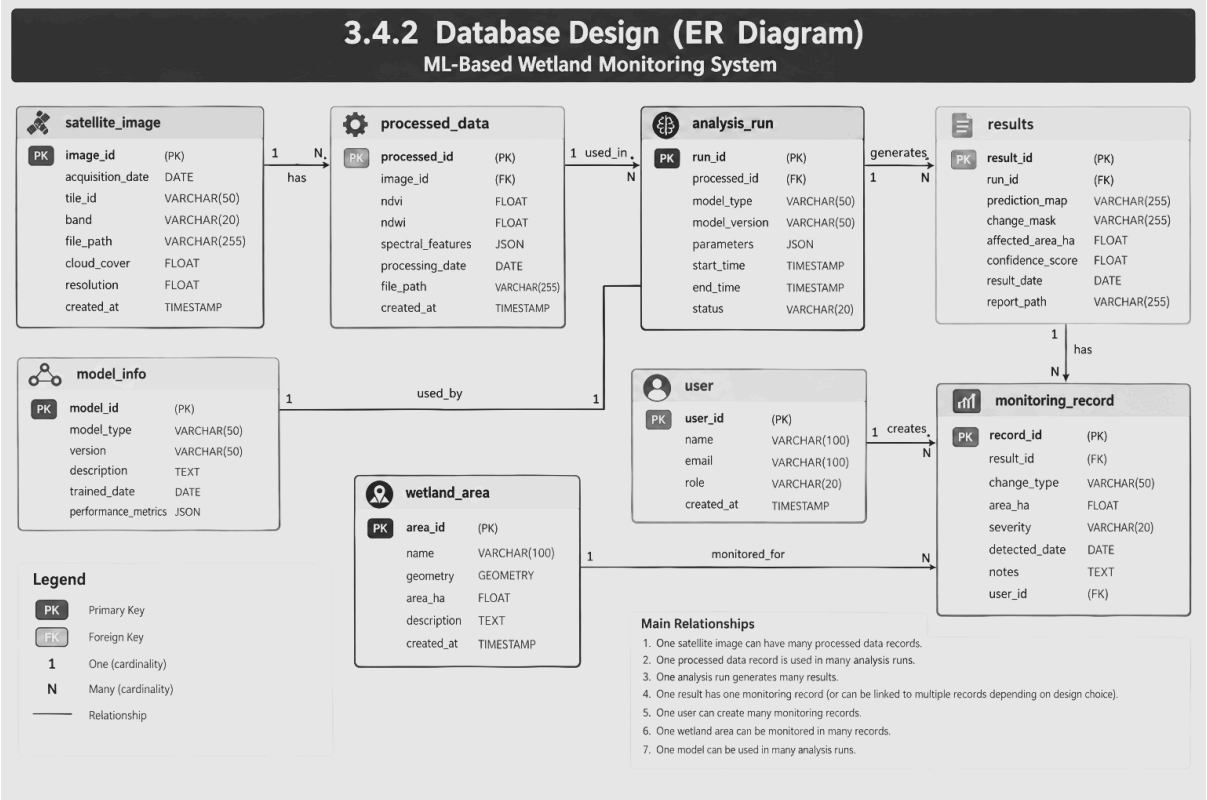
1. **Architecture Diagram**

The architecture diagram illustrates the overall structure of the proposed wetland monitoring system and the interaction between its main components. It shows how satellite imagery and other relevant data are collected, processed, analysed using machine learning, and transformed into detected, quantified, and mapped wetland changes for monitoring and interpretation.



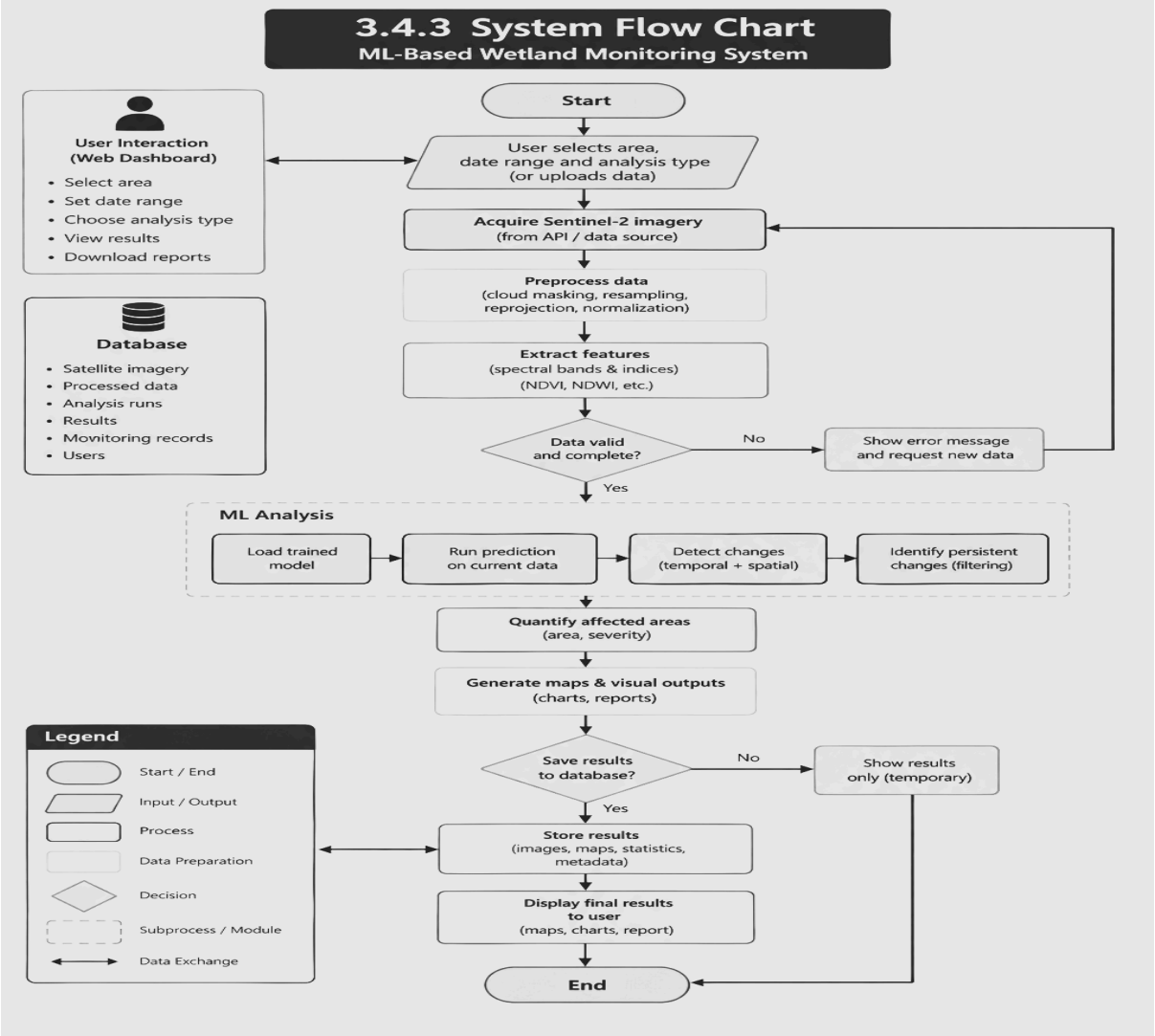
3.4.2 Database Design

The database will store satellite imagery metadata, processed data, machine learning analysis runs, detected changes, monitoring records, model information, wetland areas, and user information. The relationships between these entities will allow the system to maintain and retrieve data required for wetland monitoring, analysis, and reporting.



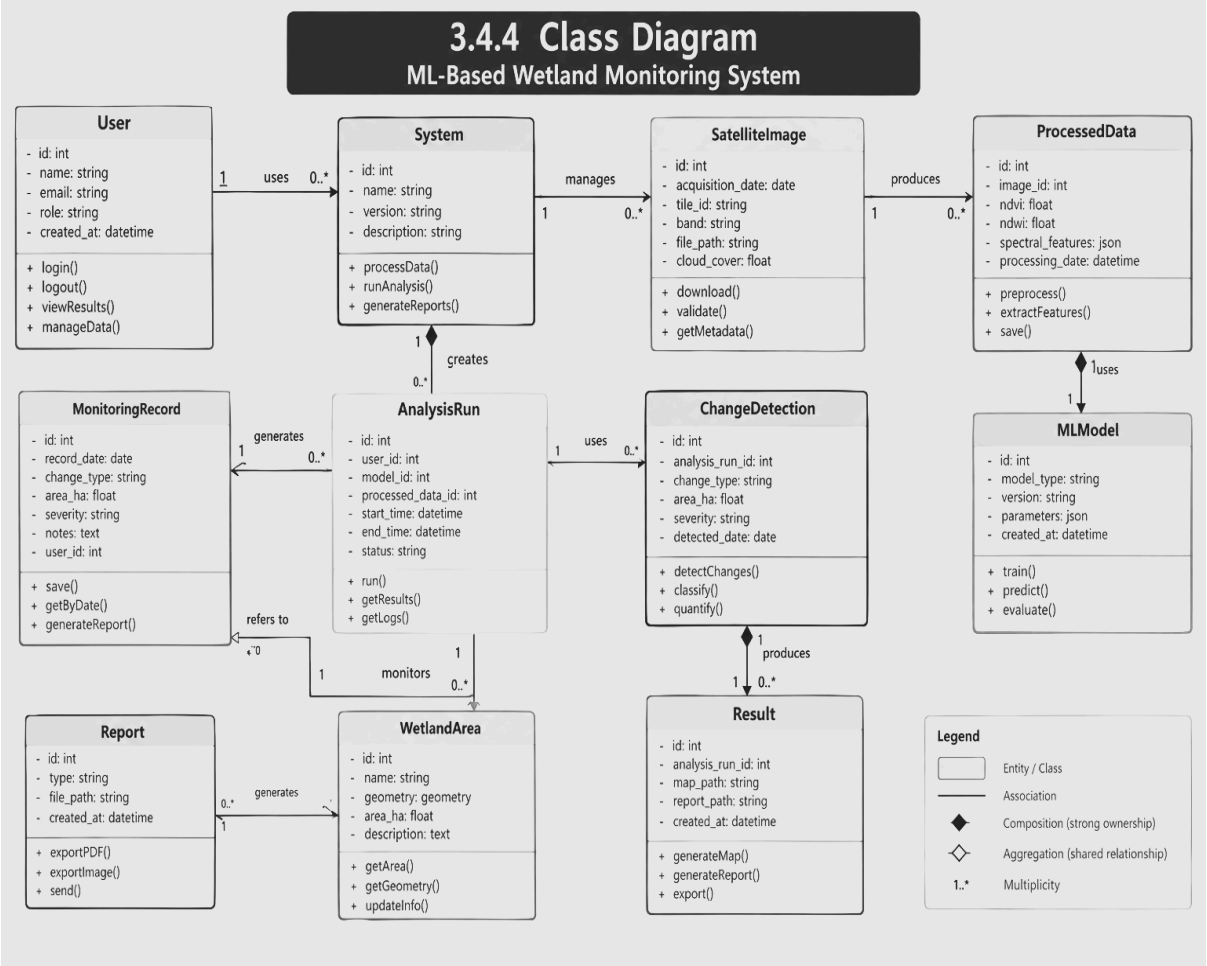
3.4.3 System Flow Chart

The system flow chart presents the sequence of processes followed by the proposed system, from data acquisition and preprocessing through feature extraction and machine learning analysis to change detection, quantification, mapping, and presentation of results. It provides a step-by-step view of how the system processes input data to produce monitoring outputs.



3.4.4 Class Diagram

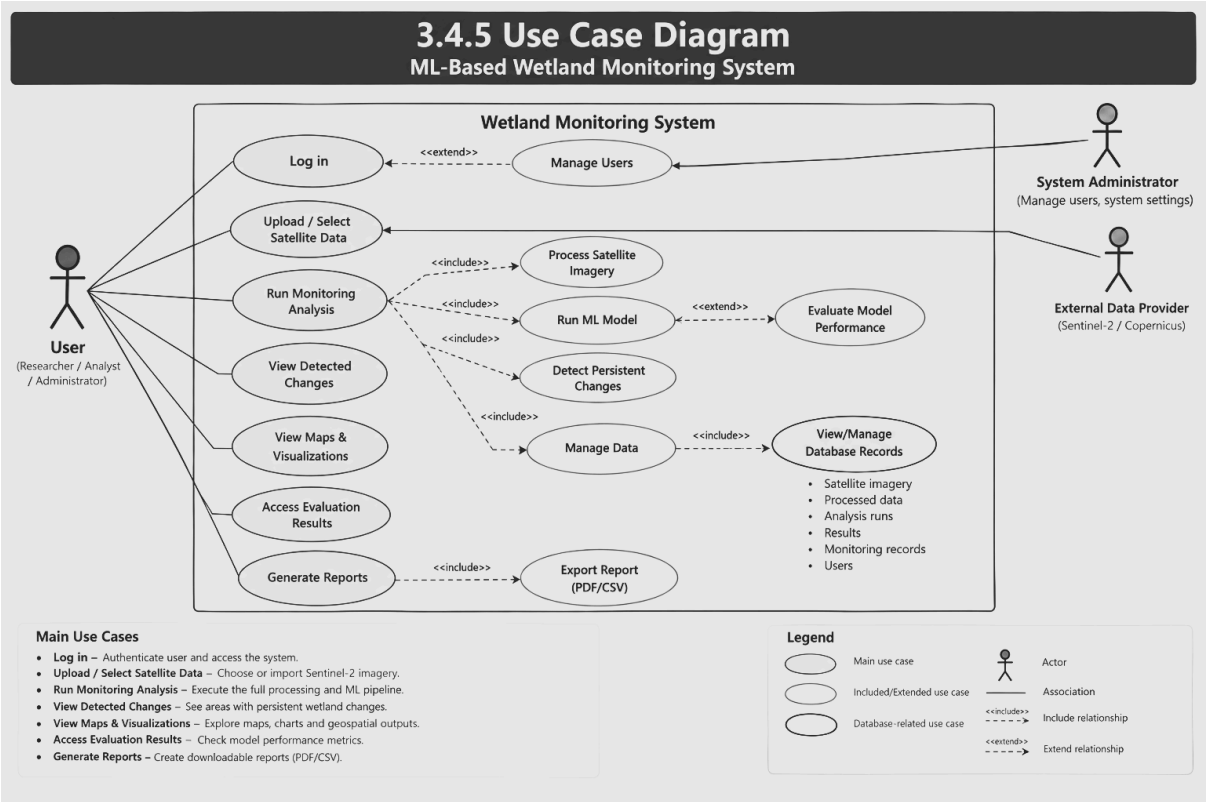
The class diagram represents the main classes of the proposed system and the relationships between them. It describes the structure of the software by showing the key entities, their attributes, methods, and interactions required to support data management, image processing, machine learning analysis, change detection, and result visualization.



3.4.5 Use Case Diagram

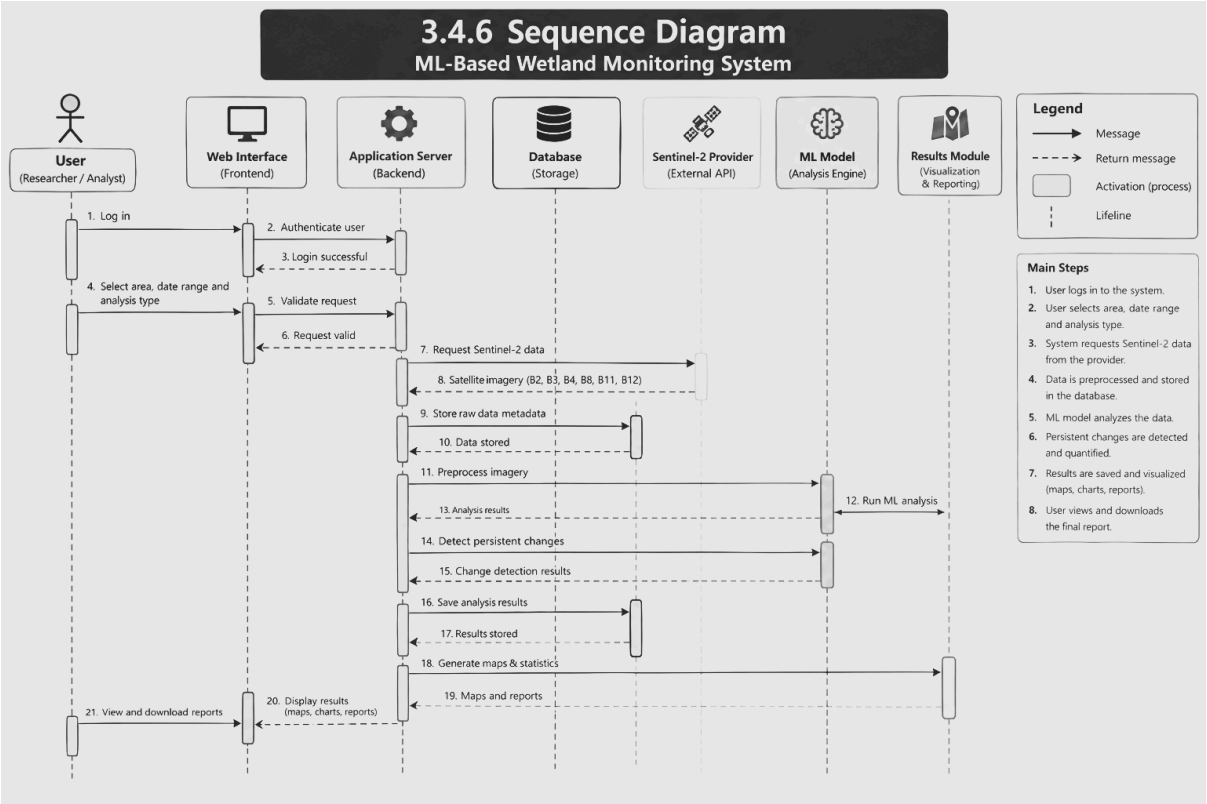
This diagram shows how the main user, system administrator, and external satellite-data provider interact with the wetland monitoring system.

The main user activities are selecting satellite data, running monitoring analysis, viewing detected changes and maps, accessing evaluation results, and generating reports. The internal use cases show the processing and ML operations required to perform those activities.



3.4.6 Sequence Diagram

This sequence diagram shows the interaction between the user, web interface, application server, database, Sentinel-2 provider, ML model, and results module during a monitoring operation.



3.5 Development Tools

The following tools and technologies will be used to develop the proposed wetland monitoring system:

- Python: Main programming language for data processing and machine learning.
- Google Earth Engine: For accessing and processing multi-temporal Sentinel-2 satellite imagery.
- Pandas and NumPy: For data manipulation and numerical processing.
- Rasterio and GeoPandas: For processing and analysing geospatial and raster data.
- Scikit-learn: For developing and evaluating machine learning models.
- Flask: For developing the backend and connecting the ML model to the web application.
- PostgreSQL/PostGIS: For storing and managing spatial monitoring data.
- Leaflet: For displaying detected changes and monitoring results on interactive maps.
- HTML, CSS and JavaScript: For developing the web interface.
- Git and GitHub: For version control and project management.

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